Our IEEE Paper Explained Simply

"Skin Cancer Detection Using Deep Learning: A Comparative Analysis and a Multi-Model CNN–Transformer Ensemble with External Validation"

Team guide — read this with `sc_two_column.pdf` open next to it

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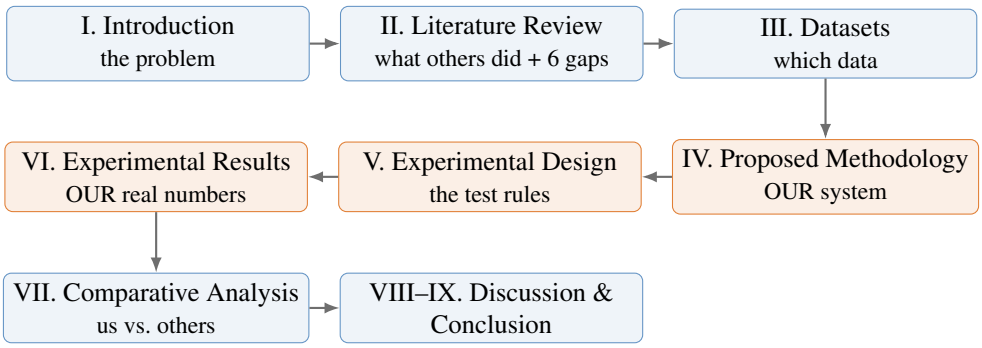
1 The Whole Paper in One Minute

- **Topic:** using AI to look at a photo of a skin spot and say whether it may be skin cancer.
- **Problem we found:** many papers claim 95–99% accuracy, but these numbers are often too good to be true. The same photo (or the same mole) appears in both training and testing, models are never tested on photos from other hospitals, and they work worse on dark skin.
- **What we did:**
  1. Read and summarized 40+ research papers (literature review).
  2. Studied 12 public datasets and chose the best two.
  3. Designed our system (proposed methodology) and an honest testing plan (experimental design).
  4. **Trained 3 models** — EfficientNetV2-S and ConvNeXt-T (CNNs) and Swin-T (transformer) — on ISIC 2019 and **combined their votes** (our USP).
  5. Compared everything with 11 kinds of published algorithms (comparative analysis).
- **Our main result:** on 3,596 test images the combined model reached **77.95% accuracy**, **62.99% balanced accuracy**, AUC **0.950**, and caught **95.4%** of cancers — better than any single model or any pair of models.
- **Honest finding:** on phone photos from another country (PAD-UFES-20) accuracy fell to 37%. *How* and *where* you test a model matters more than *which* model you choose.

In simple words: The paper says: "Here is what everyone else has done, here is what is still wrong, here is the best data, here is our 3-model system, here are our real results, and here is how they compare with others."

How the sections connect



2 Title, Abstract, and Keywords

Where in the paper: page 1

Title. "Skin Cancer Detection Using Deep Learning" is our project name. The subtitle names our two contributions:

- **Comparative analysis** = we compare many published algorithms fairly.
- **Multi-model CNN–transformer ensemble** = two CNNs and one transformer vote together (our USP).
- **External validation** = we also test on a completely different dataset (phone photos from Brazil).

Abstract. A one-paragraph summary: why early detection matters, why published results are hard to compare, what we reviewed, which models we trained, and our main numbers (77.95% accuracy, 62.99% balanced accuracy, AUC 0.950 internally; 37.25% accuracy but best cancer AUC 0.873 on phone photos).

Keywords. Ten words that help search engines (IEEE Xplore) find the paper.

3 Section I — Introduction

Where in the paper: pages 1–2

Paragraph 1 — why it matters. Skin cancer is very common. Melanoma is the deadliest; BCC and SCC are the most common. Finding them early saves lives.

Paragraph 2 — AI can help. In 2017 an AI matched dermatologists [Esteva, *Nature*]. A 2024 summary of 53 studies found AI catches 87% of cancers and correctly clears 77% of harmless spots.

Paragraph 3 — but there are two problems. (1) Results are hard to compare and often inflated by duplicate photos and random splits. (2) Good lab results do not always work in hospitals or on dark skin.

Paragraph 4 — our 6 contributions. Literature review, dataset analysis, proposed method, honest test plan, **measured results**, and comparative analysis.

In simple words: "AI for skin cancer looks great on paper but is often tested unfairly. We built a 3-model system and tested it fairly."

4 Section II — Literature Review

Where in the paper: pages 2–4, Table I

This section summarizes what other researchers have done, in 7 topics plus a list of gaps.

Subsection	What it says in simple words
A. Early deep learning	2017–2019: AI matched or beat skin doctors in lab tests, but the photos were hand-picked.
B. CNNs	Most papers take a CNN pre-trained on everyday images and teach it skin photos. Results: 83–93% on HAM10000.
C. Transformers, hybrids, ensembles	Transformers look at the whole picture. Mixing CNN + transformer works better than either alone. Teams of models win competitions.
D. Segmentation	U-Net and its upgrades draw the outline of the lesion.
E. Imbalance & synthetic data	Cancer photos are rare. *Focal loss* helps (+9.5 points in one study).
F. Explainability & foundation models	Heatmaps (Grad-CAM) show where the AI looked. Huge "foundation models" like PanDerm are strong but need big computers.
G. Clinical validation & fairness	Real hospital studies: AI catches most cancers but gives many false alarms. Dark skin is badly represented.

H. Research Gaps (the most important part of Section II)

Gap	What our project did about it
G1: Unfair, incomparable testing	Removed duplicate photos (344 groups merged), kept each mole in one split, used balanced accuracy
G2: Never tested on outside data	Tested on PAD-UFES-20 (phone photos), never used in training
G3: Rare cancers (SCC, AK) ignored	Trained on all 8 classes with focal loss; reported recall per class
G4: Skin-colour bias	Reported results per Fitzpatrick skin type (DDI planned for dark skin)
G5: Heatmaps not checked	Grad-CAM module planned for the next stage
G6: Best models too heavy	All 3 models trained on a laptop GPU (6 GB) in under 1 hour each

5 Section III — Datasets

Where in the paper: pages 3–4, Table II, Fig. 1, Fig. 2

We looked at 12 datasets and used the **two best**:

- **ISIC 2019** (25,331 photos, 8 classes) — split into training 18,076, validation 3,659, internal test 3,596.
- **PAD-UFES-20** (2,298 phone photos) — external test only.

The others (MILK10k, DDI, ISIC 2018 masks, PH2, ISIC 2020) are marked "Future" in **Table II**.

Fig. 1: harmless moles (NV) have 12,875 photos but SCC only 628 — this is why we use focal loss. **Fig. 2**: the share of cancer images ranges from 0.1% to 47% between datasets — this is why plain accuracy can mislead.

6 Section IV — Proposed Methodology (OUR System)

Where in the paper: pages 4–6, Fig. 3, Eqs. (1)–(5), Algorithm 1, Table III

Fig. 3 shows the full system. Subsection A says clearly which parts we have *already trained* and which come next.

Subsection	What happens (simple words)
A. Implemented configuration	What is done now: the 3 models + voting. Segmentation, patient details, and heatmaps are the next stage.
B. Preprocessing	Resize photos (shorter side 256 px), random 224×224 crops for training.
C. Segmentation (next stage)	Attention U-Net will cut out the lesion.
D. Hybrid backbone ensemble	EfficientNetV2-S and ConvNeXt-T (CNNs: texture and colour) and Swin-T (transformer: overall shape).
E. Metadata fusion (next stage)	Age, sex, and body part will be added.
F. Imbalance-aware training	Focal loss + extra weight for rare classes + flips, rotations, colour changes.
G. Calibrated ensemble	Each model's confidence is calibrated, then the 3 models vote (equal weights, or weights learned on validation). The referral threshold catches at least 95% of cancers on validation.
H. Explainability (next stage)	Grad-CAM heatmap of where the model looked.
I. Training procedure	Algorithm 1 (recipe) and Table III (exact settings, laptop GPU, 36–49 min per model).

The equations in simple words

Equation	Meaning
(1) $\mathscr{L}_{seg}$	Score for training the outline model (overlap of predicted and true outline).
(2) $\mathbf{z}_m, \mathbf{p}_m$	Join image features with patient details, then turn them into class probabilities.
(3) $\mathscr{L}_{FL}$ (focal loss)	Error score multiplied by $(1-p)^{\gamma}$: easy photos barely count, hard or rare photos count a lot.
(4) $\hat{\mathbf{p}}$ (ensemble)	Final answer = weighted average of the 3 calibrated models, each averaged over 4 flipped copies of the photo.
(5) Grad-CAM	Colours the parts of the image that most increased the predicted class score.
(6) BA	Balanced accuracy: average of per-class recall.
(7) Spec, F1	Specificity (harmless spots correctly cleared) and F1 (balance of catching cases and avoiding false alarms).

7 Section V — Experimental Design (The Test Rules)

Where in the paper: page 6

- **A. Leakage-free protocol:** duplicates removed, photos of one mole kept together, the test sets used only once at the end.
- **B. Metrics:** accuracy, balanced accuracy (main score), F1, AUC, cancer sensitivity/specificity, calibration error, and 95% confidence intervals.
- **C. Fairness:** results per Fitzpatrick skin type on PAD-UFES-20.
- **D. Ablation:** every combination of the 3 models is tested to show how much combining helps.

8 Section VI — Experimental Results (OUR Real Numbers)

Where in the paper: pages 6–9, Figs. 4–7, Tables IV–VII

Part	What it shows (simple words)
Fig. 4 — training curves	Models learned fast in the first 5 epochs; then validation stopped improving (overfitting), so we kept the best epoch.
Table IV + Fig. 5(a)	Internal test: single models 73.1–76.9% accuracy; **ensemble 77.95%**, balanced accuracy 62.99%, AUC 0.950, 95.4% cancers caught.
Fig. 6 — confusion matrix	Counts of right and wrong answers: 2,803 of 3,596 correct (diagonal). Best: nevus 91%, vascular 81%, BCC 80%, melanoma 66%. Hardest: SCC 49%, AK 43%, DF 39%.
Table V — per class	Precision, recall, and F1 for each class; macro-precision 68.7%.
Fig. 7 — ROC curve	The ensemble separates cancer from non-cancer best (AUC 0.934).
Table VI — ablation	**USP proof:** 1 model $<$ 2 models $<$ 3 models on every column.
Table VII + Fig. 5(b)	External phone photos: accuracy only 33.7–37.3%, but the ensemble still has the best cancer AUC (0.873).
Skin types	Similar results for skin types I–II and III–IV; only 11 photos of types V–VI, too few to judge.

In simple words: "Combining 3 models works better than any one model. But a model trained on close-up dermoscope photos does not yet work well on normal phone photos."

9 Section VII — Comparative Analysis of Algorithms

Where in the paper: pages 8–12, Table VIII, Figs. 8–12, Table IX

We compare published algorithms in **4 fair groups (panels)** and add our own result.

Part	What it shows
Table VIII	All published results + our ensemble (bold row in Panel B).
Fig. 8 — Panel A	11 algorithms on HAM10000: SVM 53%; CNNs 74–93%; transformers/hybrids up to 94% (random splits, optimistic).
Fig. 9 — Panel B	ISIC 2019: the competition winner scored 63.6% on its hidden test; **ours 63.0%** on our held-out test (orange bar).
Fig. 10 — Panel C	Skin-specific pre-training (PanDerm) is best.
Fig. 11 — Panel D	Attention U-Net is the best practical outline model.
Fig. 12 — Clinic	In hospitals, AI catches cancers but gives many false alarms.
Table IX	Pros and cons of each algorithm family.

10 Sections VIII–IX — Discussion and Conclusion

Where in the paper: pages 11–12; references pages 12–14

- **Why does the ensemble help?** Each model makes different mistakes (e.g., Swin-T is best at AK, ConvNeXt-T best at BCC); averaging fixes some of them.

- **Honest limitations:** one data split, 224 px images, no segmentation/metadata yet, learned weights did not beat equal weights, phone-photo accuracy is low, too few dark-skin images.
- **Conclusion:** combining multiple algorithms reliably improves detection (77.95% accuracy, 62.99% BA, AUC 0.950); transfer to new imaging conditions is the main open problem.
- **Future work:** add segmentation, metadata, Grad-CAM; higher resolution; fine-tune on phone photos; test on DDI; phone app.

11 Important Words (Glossary)

Word	Meaning
Dermoscopy	Close-up skin photo taken with a magnifying device
Accuracy	Share of all photos classified correctly
Balanced accuracy	Average accuracy per class, fair to rare classes
Sensitivity	Of all real cancers, the share the model catches
Specificity	Of all harmless spots, the share the model correctly clears
AUC	0.5 = guessing, 1.0 = perfect separation of classes
Ensemble / soft voting	Averaging the probabilities of several models
Data leakage	Test information sneaking into training (scores look too good)
External test	Testing on a completely different dataset
Epoch	One pass of the model over all training photos
Overfitting	The model memorizes training photos instead of learning general rules
Calibration	Making the model's confidence match reality
TTA	Predict on flipped copies of the photo and average
Ablation study	Adding or removing one part at a time to see how much it helps
Confidence interval	The range in which the true score probably lies
Fitzpatrick skin type	Scale from I (very light) to VI (very dark)

12 Questions Your Teacher or Examiner May Ask

Q1. What is your USP? We combine three different algorithms (two CNNs and one transformer). Our ablation (Table VI) proves that the combination beats every single model and every pair.

Q2. What accuracy did you get? 77.95% accuracy, 62.99% balanced accuracy, macro-AUC 0.950, and 95.4% cancer sensitivity on 3,596 held-out ISIC 2019 images.

Q3. Why is balanced accuracy lower than accuracy? Most photos are harmless moles, which are easy. Balanced accuracy gives rare cancers equal weight, so it is the fairer (and official ISIC 2019) score.

Q4. Why not 95–99% like other papers? Those papers usually split photos of the same mole into train and test. We prevent that, so our numbers are honest. The ISIC 2019 winner scored 63.6% on its hidden test; we scored 63.0%.

Q5. Why did accuracy drop on PAD-UFES-20? Our models learned from dermoscope close-ups; PAD-UFES-20 has normal phone photos and a different class mix. This is a known real-world problem.

Q6. Why focal loss? It makes the model pay attention to rare cancers instead of only common moles.

Q7. Why did equal weights beat learned weights? The learned weights fitted the validation set too closely; simple averaging generalized better.

Q8. How long did training take? 43, 49, and 36 minutes on an RTX 3060 laptop GPU (about 2 hours including evaluation).

Q9. Can it replace a doctor? No. It is a decision-support tool; it suggests referral and a doctor decides.

Q10. What is next? Segmentation, patient details, Grad-CAM, higher resolution, fine-tuning on phone photos, and testing on DDI.

Tip: read this guide with `sc_two_column.pdf` open. For all measured numbers and graphs, see `Training_Results_Report.pdf`.